

Driver Drowsiness Detection - A Snippet of ADAS Using OpenCV

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Abstract—These days, driving while fatigued is a serious problem. Drivers of buses, trucks, and cars travel great distances both during the day and at night. Accidents and losses on the road are mostly caused by driving, even while fatigued. Establishing a mechanism to track an individual's level of tiredness and its manifestation is crucial. A behavioral system is the one that was created. A method for supervised learning is utilized to identify tiredness. The current method uses a camera to capture the footage, detects the driver's face frontal characteristics (such as their lips and eyes), and applies machine learning algorithms to each frame. The crucial figures determining sleepiness are the eye aspect ratio (EAR) and mouth aspect ratio (MAR), which can be identified with the help of the OpenCV library for facial recognition. This targets the driver's visible facial characteristics. The OpenCV can recognize a person's face with ease, and if the driver exhibits even the slightest hint of sleepiness, the individual is alerted. This contributes to a reduction in traffic accidents, which in turn saves many lives.

Keywords—Image Processing, Real time video surveillance, Drowsy Driver Detection, Alarm, lighting conditions.

I. INTRODUCTION

Driver fatigue constitutes a significant and widespread factor in a wide range of road accidents. Recent statistics and research findings indicate that numerous fatalities and injuries annually can be attributed to accidents linked to fatigue. Globally, road accidents result in substantial loss of life and well-being, with approximately 2,400 road accidents occurring daily, translating to one fatality every four hours. It has been estimated that around 20% of vehicle accidents involving driver fatalities are a direct consequence of driver drowsiness. Moreover, it's been observed that a rapid decline in a driver's performance due to increased tiredness is responsible for over 20% of all automobile accidents. Drowsiness adversely impacts a driver's attentiveness, leading to distraction and an increased likelihood of road accidents. Injuries associated with drowsiness tend to be more severe due to the higher speeds involved and the incapacitation of the driver to take evasive actions or apply brakes before a collision occurs. Therefore, developing solutions for detecting or preventing driver fatigue represents a crucial challenge within the realm of accident prevention systems. Given the significant threat that drowsiness poses on the road, it is imperative to devise strategies for monitoring its effects. The diminished attention resulting from fatigue leads to various changes in a person's physiology and behavior. These alterations and parameters allow us to effectively gauge the level of drowsiness.



Fig.1.Facial Recognition

Fig 1 show in Facial Recognition Various methods for identifying drowsiness can be categorized into two primary groups. The methods in the first group determine the degree of fatigue by assessing physiological changes within the body.[1] Examples of such methods include monitoring eye condition, speech characteristics, and the time interval between yawns, head position, seating posture, heart rate, and brain signals. [2] Drowsiness also causes noticeable alterations in driving behaviour. The methods belonging to the second classification determine a driver's level of drowsiness by monitoring said transformations. Parameters such as steering angle, following distance, lateral vehicle position, longitudinal speed, longitudinal acceleration, and lane deviation are employed in the methods belonging to the second category. These techniques aid in assessing a driver's level of drowsiness and play a crucial role in promoting road safety.

II. RELATED WORKS

This research delved into the synchronisation of dynamic actions, specifically complexity, within electrocardiogram (ECG) and electroencephalography (EEG) measurements related to coronary heart rate and brain activity. Sleeping-deprived participants were exposed to real-life circumstances for data collection purposes. The findings reveal a plausible correlation between the two physiological systems as synchronous reductions in intricacies indicate an interaction [3].

Preliminary results obtained from examining four subjects suggest the existence of a slight to moderate association between these biological oscillations. To assess both heart rate and brain signals, we calculated power spectra and investigated the synchronization patterns with common frequencies in both systems [4].

From 1989 to 1993, the Regulatory Body responsible for vehicle safety recorded that a yearly count of accidents on American roads had resulted from wheelman's

exhaustion or drowsiness. In order to determine the frequency of accidents and how driving ability is affected by tiredness, the latest research looked at the effects of extended sleep deprivation. Given the dangers of driving in actual traffic, a realistic highway driving simulator was used. A group of twelve individuals, ranging from 26 to 35 years old, underwent driving performance evaluations through a range of techniques including ongoing electroencephalogram (EEG) observation, video recordings, and review of both their driving data and completed questionnaires.

From 1989 until 1993, the Vehicle Safety Regulatory Body reported an annual incidence of car collisions on American highways caused by sleep-induced impairment or droopiness. In order to determine the frequency of accidents and how driving ability is affected by tiredness, a new study looked at the effects of extended sleep deprivation. A high-fidelity highway driving simulator was used as actual driving circumstances carry dangers. To assess the impact of sleep deprivation on driving ability, twelve participants aged 26-35 were subjected to several assessments such as EEG monitoring, video recording, and analysis of both driving performance data and questionnaires. The results showed that after being awake for 36 hours (on Day 3) or a staggering 60 hours (on Day 4), there was an alarming increase in accidents and lane deviations which posed significant risks to highway safety. Sleep deprivation was also linked to less obvious traffic safety indicators, such as speed and lateral positioning fluctuations [5].

Despite the emergence of some tendencies, partial sleep deprivation did not significantly alter any of the variables; this may have been because the individuals were young, in excellent health, did not use medicine, and did not have any sleep debt at the beginning of the investigation. Utilizing the gathered data, a first neural network analysis is presently being conducted. Identification of driving performance trends might be a compelling argument in favor of developing an in-car neural network system to identify and alert fatigued drivers to possible hazards.

Another investigation focused on the impact driving if slightly exhausted time impact future attention along with efficiency among vehicle chauffeurs. A total of twenty young men people, aged twenty-five to fifty-five, with no history of sleep difficulties, took into account virtual vehicle activities from two and four evening. In one sitting, the attendees were deprived of sleep and permitted to rest only between 3 AM and 7 AM on the prior night. Their driving performance was monitored alongside EEG readings and KSS ratings throughout the task. The research revealed that although all parameters were affected by drive time, lack of sleep impacted KSS scores but not in alpha-theta spectral power. This effect was magnified through restricted sleeping patterns. Specifically, while functionality decreased significantly with longer drives, only one of several analysed performance indices - right-hand edge crossings- varied due to limited rest periods during this study period. The association between alertness and decline in overall performance is examined in light of these findings [6].

Additionally, Template Matching, a method for locating a template image within a larger image, was discussed. The `cv.matchTemplate()` function offered by OpenCV is used to slide the template image across the input image and determine its level of similarity at each position. Several evaluation methods are implemented in OpenCV, and the result is a grayscale image indicating the extent of similarity between the template and the input image's local patches.

Three unique steps are involved in a productive approach to sleepiness detection, as demonstrated in the study by Manu (2016) [7]. These include eye tracking, yawn detection, and facial feature identification using the Viola Jones approach. After identifying the face, the algorithm isolates the skin region and filters out background features that aren't faces based only on skin tone. This normalizes lighting conditions. Matching templates using the correlation coefficient are employed to detect yawns and track eye movements.

Belal et al. (2013) [8] introduced a module for the Advanced Driver Assistance System (ADAS), aimed at enhancing transportation safety and reducing accidents caused by driver fatigue. This technology makes use of artificial intelligence and optical data to automatically identify driver sleepiness. A dashboard-mounted camera is used in a unique tiredness detection method that was developed by Jang et al. (2018) [9]. The face of the driver in the photo is recognised by the computer, which then provides an estimation of facial landmarks. The methodology employed relies on utilising an AdaBoost classifier that depends upon features derived from Modified Census Transform to distinguish and identify faces. A continuous sleepiness detection method that considers individual driver variations was developed by Feng et al. (2019) [10]. A Space Invariant Artificial Neural Network is utilised to identify the driver's facial region, thereby resolving any issues of low precision that may arise due to attrition of characteristics. To detect the frontal facial characteristics of the driver, one can make use of Dlib toolbox in locating its respective landmarks. An additional parameter, Eyes Aspect Ratio, is presented to evaluate driver fatigue depends on eye landmarks.

In 2018, Anirban et al. [11] recommended a smartphone-based solution for identifying driver sleepiness. A three-stage sleepiness recognition technique is used by the system. Using an adjusted eye state classification algorithm, the front camera captures pictures that are used in the first step to determine the Percentage of Eyelid Closure (PERCLOS). While driving in the nighttime, near-infrared flares shine on the driver's face. In stage two of analysis, data obtained through microphones is used to determine the Sonorous-to-Nonsonorous ratio. A final verification stage is employed to confirm the driver's drowsiness within a specified time frame and trigger an alert.

The intent of Igor et al. (2019) [12] was to identify driver tiredness and attention states quickly by employing the forward-facing camera of smartphones to continually analyse driving face characteristics. There are two types of emergency state recognition: online and offline. Computer vision libraries such as OpenCV and Dlib were utilised to

recognise dangerous driving conditions in real-time while driving, made possible by the effectiveness and efficacy of smartphones.

The Histograms of Oriented Gradients, which were employed for presenting image edge data, were enhanced by Shibo and Xiaojie (2013) [13]. To fulfil the real-time tracking demands and attain the necessary accuracy, they integrated Histograms of Oriented Gradients with background subtraction detection. Ahmad et al. (2019) [14] carried out a thorough investigation into the effects of being sleepy while operating a vehicle. The social, physiological, and neurological changes which take place during alertness and drowsiness, as well as the symptoms, causes, accident statistics, and sleep stages are all covered in this analysis. It also takes into account social information from drivers, different analytical methods, biomedical indicators of human health, and their application in detecting drowsiness. Wei et al. (2012)

[15] introduced a non-intrusive drowsiness identification approach utilising pupil-tracer and virtual data analyser. They developed a strong iris recognition mechanism to tackle concerns related to changes in lighting and driver posture.

III. PROPOSED METHOD

The OpenCV Classifier technique is used in the proposed driver sleepiness recognition technique to analyze and identify the driver's face and eyes from pictures taken with a camera. This technology performs well even in less than ideal lighting, correctly identifying the driver's eyes and facial characteristics in a range of lighting situations. After identifying the car operator's visage and orbs, the system proceeds to calculate their EAR by analysing both vertical and horizontal distances between them. Additionally, it determines their MAR by measuring equivalent distances around the mouth area. Fig.2. show in Showing the process of EAR.

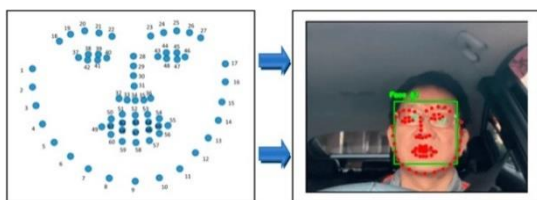


Fig.2. Showing the process of EAR

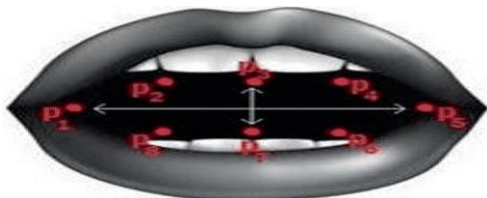


Fig.3. Depicts MAR

Fig.3. show in Depicts MAR. The system uses these computed figures to determine the level of drowsiness in the Wheelman. Extensive research has been conducted in the past to analyze the effectiveness of criteria for detecting driver drowsiness. One of the system's primary

features is its seamless integration into automobiles for real-time driver drowsiness monitoring. Its performance continuously demonstrates its capacity to identify driver fatigue correctly and dependably, therefore making a substantial contribution to the reduction of accidents. When the technology was first conceived, a camera was used to capture the facial image of driver. Fig.4. Show in Proposed Method.

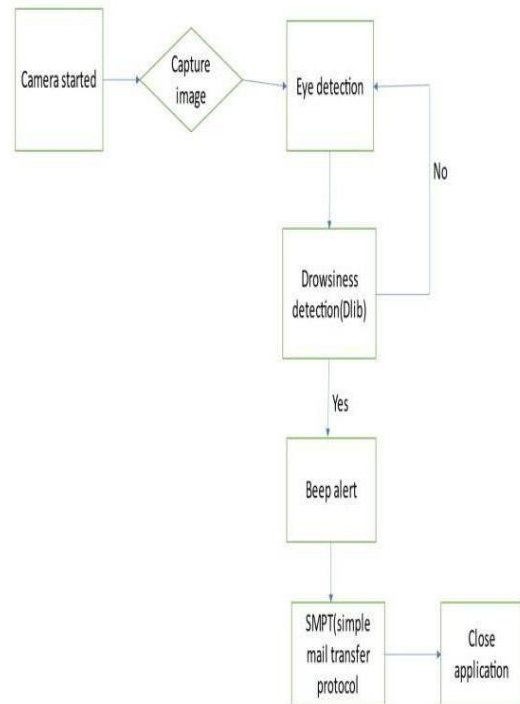


Fig.4. Proposed Method

IV. RESULT & DISCUSSION

The development and evaluation of the driver drowsiness recognition approach involved the utilization of the OpenCV library and the Python programming language. To assess the system's effectiveness, a comprehensive dataset comprising images and videos depicting drivers across various races, age groups, and genders in different states of drowsiness was employed. This dataset was thoughtfully compiled from diverse sources.

The results of the experiments showcased the system's remarkable ability to accurately define driver drowsiness with a high degree of precision. It excelled in recognizing both positive and negative instances of drowsiness, as indicated by its impressive precision and recall metrics. Furthermore, the system's real-time capabilities were put to the test using a live video stream from a car's camera.

Significantly, the system exhibited its proficiency by promptly detecting driver drowsiness in real time. Notably, variations in lighting conditions or changes in the driver's head position had no adverse impact on the system's performance, as the experiments conclusively demonstrated. It remained effective even when faced with scenarios where the driver's head was turned or tilted, consistently and accurately identifying the driver's facial features.

The real-time drowsiness detection was done in this research to explore the effectiveness and functionality of the proposed technique. In case where the selected image is facing illumination, noise, and view-point variation, the recommended technique is quite enough to manage and perform prediction better than others. For this analysis, sample images were trained and tested to visualize the result, and the outcome is satisfactory, which is far better than others. The following Fig 6 depicts the flow analysis followed here and the steps carried out within this entire work to predict drowsiness.

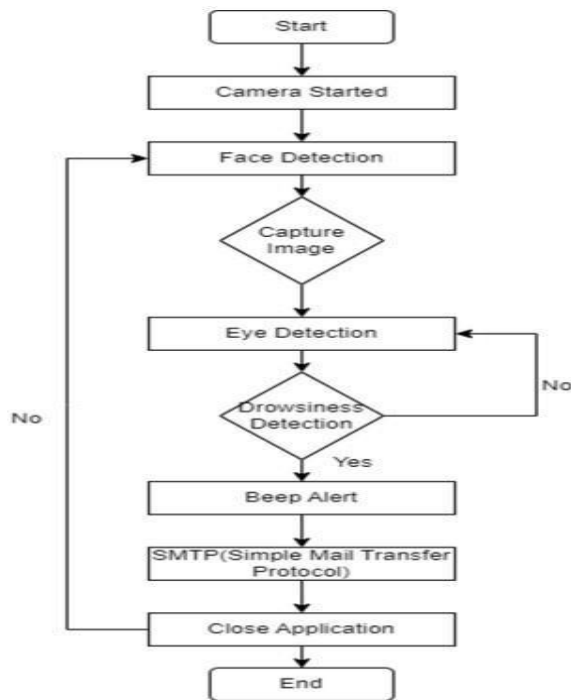


Fig.5.Flow analysis of steps involved in it

Also, figures 6 and 7 represent the images classified as per the procedure enclosed in figure 6 to classify them as having drowsiness or without.



Fig.6.With Drowsiness



Fig.7.Without Drowsiness

V.CONCLUSION

Based on the insights gathered from various research studies and academic papers, it is evident that a driver's drowsiness detection system holds significant importance in enhancing road safety. Considering the findings, it is reasonable to advocate for the integration of such a system as a mandatory component of a driver's toolkit. This would contribute to reducing accidents caused by driver fatigue and promote safer driving practices.

To put this into action, it's essential to design and enhance the implementation of a Driver Drowsiness Detector using Python and OpenCV, coupled with a webcam for capturing facial data. Such a system can provide real-time monitoring and alerts to prevent accidents resulting from drowsy driving. The integration of computer vision and machine learning techniques, as described in the referenced studies, can contribute to the effectiveness of such a system. This proactive approach to road safety can make a significant difference in preventing accidents and saving lives.

A real-time image-based drowsiness identification method was presented in this study. A camera was utilized to identify the driver in real time and extract the indicators of fatigue from the lips, head, and eyes in order to execute drowsiness detection. Finally, the suggested strategy was put into practice. An alert goes off, warning the motorist and guaranteeing road safety, when a sleepiness indication is spotted. To create a mobile application in the future so that consumers may conveniently operate the system while driving.

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